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Description - Contains theory for the PSG portion of IPG and IPG II Boards.

IPG Pulse Sequence Generator (PSG)

The Pulse Sequence Generator (PSG) section of the IPG board generates digital waveforms that are played out to the Gradient Amplifier, to the Combined Receiver Exciter DAB (Data Acquisition Board)(CERD), and to the Sequence Synchronous Protocol (SSP) bus. These digital waveforms are played out of a common bank of static RAM which is 256K by 16 bits. The method of addressing this common waveform memory is controlled by XILINX Field Programmable Gate Arrays, or Logic Cell Arrays (LCAs) that respond to two banks of instruction memory of 32K by 32 bits each.

The SSP sequence generator latches its waveforms directly into bus drivers that drive the backplane. It can provide new values at a 1-mHz rate (once per microsecond).

There are three gradient sequence generators, x, y, and z. These generators provide a 15-bit waveform and a 16-bit scalar value to a TMS320C30 Digital Signal Processing Microprocessor which is referred to as the Waveform and Rotation Processor (WARP). Each gradient sequencer can provide new values at a 250 kHz rate (once every 4 microseconds). The WARP multiplies the waveform and scalar values. It writes a 20-bit product for each of the three sequencers to registers contained within a XILINX LCA. Each 20-bit product is then shifted out serially over the backplane at a 10 mHz rate.

There are four RF sequence generators, Rho1, Rho2, Theta1, and Theta2. These generators provide a 15-bit waveform and a 16-bit scalar to a CY7C516 multiplier. Each RF generator is capable of providing new values at a rate of 500 kHz (once every 2 μ sec). There are only two 16-bit buses on the backplane for these four sequence generators, a Rho bus and a Theta bus. The RF waveforms are multiplexed over these two busses resulting in a maximum bus activity of 1 mHz on each bus.

The timing of the playing out of all of the pulse sequence generators is controlled initially by another XILINX LCA, the Master Sequencer, which staggers their start-up times such that only one generator is attempting to access the common waveform memory at any time. Two banks of instruction memory are also shared by the sequence generators as follows: One 32K by 32 bit bank is shared by the SSP generator and the three gradient sequencers. The other 32K by 32 bit bank is shared by the four RF sequence generators.

Once started, all of the sequencers run synchronously, by themselves, with no further intervention from the master sequencer. There, unique instructions contain an End Of Sequence (EOS) bit, which stops them from playing out sequences until restarted in a staggered manner by the master sequencer.

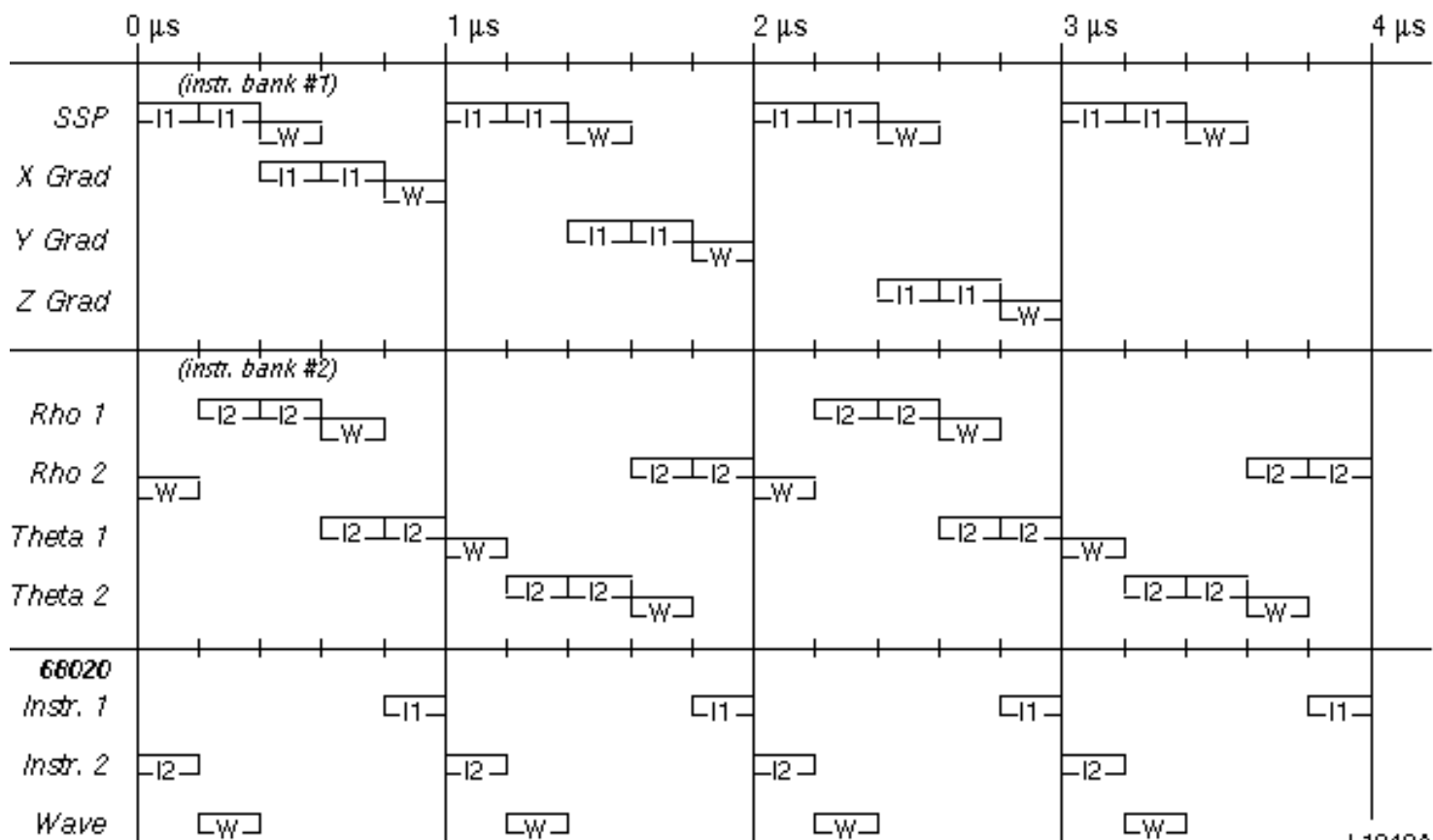
1- SEQUENCER TIMING / MULTIPLEXING OPERATION

The Master Sequencer and all of the other sequence generators (slaves) run synchronously on a 5-mHz clock. The master sequencer contains a 4- μ sec timer that runs continuously, from which all timing is derived. Each slave sequence generator has a specific time slot for which it is granted access to the shared banks of waveform and instruction memories. This timing relationship is defined graphically below in the Sequence Generator Memory Access Map (see section 2).

Once started, the slave generators generate their own latching signals to grab their specific waveform and instruction. The master sequencer controls all of the tri-state buffers, which direct the waveform and instruction memory busses to their appropriate sequence generator.

2- SEQUENCE GENERATOR MEMORY ACCESS MAP

The Sequence Generator Memory Access Map illustrates the relative location of each 200-nsec memory cycle for each sequencer. See Illustration L1949A.



SEQUENCE GENERATOR MEMORY ACCESS MAP
ILLUSTRATION L1949A

I1 blocks indicate memory cycles to instruction bank #1, shared by the SSP generator and the x, y, and z gradient sequencers. (Notice that there are no overlaps vertically for any of the blocks.)

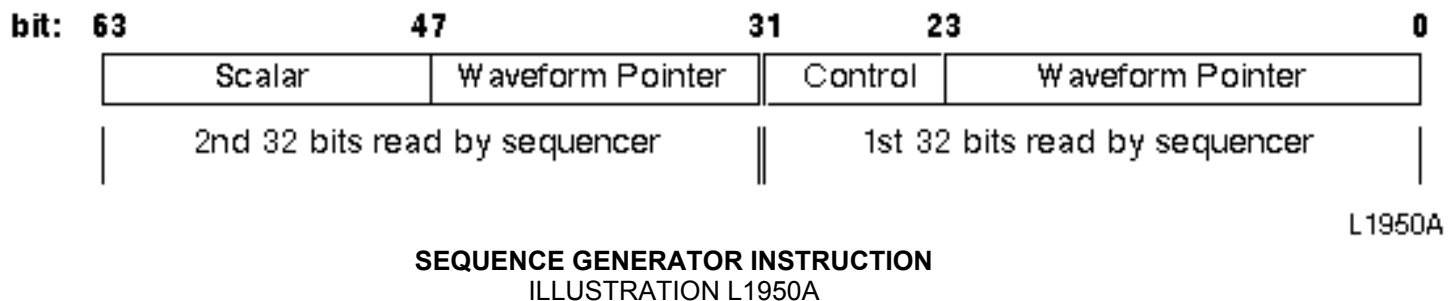
/2 blocks indicate memory cycles to instruction bank #2, shared by the four RF sequencers.

W blocks indicate memory cycles to the 256K by 16 bit waveform memory.

The 4-msec time period repeats as long as a sequence is being played out. It illustrates the maximum frequency each sequencer can operate. (Every μsec for the SSP, every 4 μsec for each gradient sequencer, and every 2 μsec for each RF sequencer.)

3- THE SEQUENCE GENERATOR INSTRUCTION

Each sequence generator responds to a 64-bit instruction containing address, period, and control bits. An instruction start address (ISA) register is located within each sequencer LCA. It is initially written to by the 68020. Instruction memory is addressed by the LCA without the offset associated with the 68020 memory map, i.e., it begins at address 0000H. It is also defined to address a 32-bit location rather than a byte address like the 68020 would use. See Illustration L1950A.



Each sequencer does two 32-bit reads of instruction memory to obtain its entire 64-bit instruction. The first 200-nsec read cycle provides the sequencer the waveform period and some control bits. During the second 200-nsec read cycle, the sequencer gets 16 bits of its the waveform address and its scalar value.

4- THE WAVEFORM PERIOD

The Waveform Period is a two's complement number of the number of "ticks" that an individual waveform point is played out. A "tick" is 1 μsec for the SSP generator, 2 μsec for the RF generators, and 4 μsec for the gradient sequencers. The value of the Waveform Period is loaded into a counter within the LCA. It determines the length of time that a particular waveform point is played out. There is only one Waveform Period for each instruction, therefore, each waveform point for that instruction plays out has the same period.

In the case of the SSP generator, referring to the Sequence Generator Access Map in Section 2, all of the waveform memory access slots will not necessarily take a value out of waveform memory. The only way a new waveform point can be read out of waveform memory for the SSP generator every microsecond is if the waveform period was set to two's complement of 1 (= FFFFFFFF). Otherwise, the SSP sequence generator waits the prescribed time before its next waveform memory access.

The gradient and RF generators act differently. They play a waveform point for every time slot, even if the waveform period counter has not expired to indicate that the next waveform address is to be driven.

5- THE WAVEFORM POINTER

The waveform pointer is the least significant 16 bits of the waveform memory address that is driven by the sequence generators. The address contained in this instruction field addresses a 16-bit location. It also starts at address 0 with no offset corresponding to the 68020 memory map. The upper two bits of waveform memory address are contained in the "control" field of instruction memory as described later.

The waveform pointer value addresses the first waveform memory point for the instruction that is being executed. The waveform memory address auto-increments once the waveform period counter has expired. The next waveform value then plays out for the length of time determined by the waveform period. This continues until an end of waveform (EOW) bit is detected in waveform memory.

5-1 Control Bits

The eight control bits (instruction bits 24 through 31) are defined as follows. There are five bits that are used only by the SSP generator. The other three are valid for all sequence generators.

bit 24: The end of sequence (EOS) bit tells the sequencer to stop executing instructions. The instruction having the EOS bit set is not played out and, therefore, the waveform period, waveform pointer, and scalar fields of the instruction are "don't cares". The other seven bits in this control field for the SSP generator are valid during an EOS instruction and are set to zero to avoid an unintended intersequence interrupt described below. When set, the EOS bit clears the waveform and scalar latches for the sequencer executing that instruction.

bits 25–27: The intersequence interrupt (ISI) bits are used only by the SSP sequence generator. There are seven possible combinations of ISI bits having at least one bit set that generates an interrupt to the 68020 CPU. The interrupt is generated within the SSP LCA. When an instruction contains at least one ISI bit set, a single, 200-nsec, active high signal is sent to a 68901 MFP chip for the 68020. The three bits are latched externally so the 68020 can determine which of the seven interrupts was generated.

bits 28–29: The waveform page bits are the most significant address bits of waveform memory. The four combinations define the four pages of waveform memory.

bit 30: The exorcist bit (SSP only) is latched externally and can be read by the 68020.

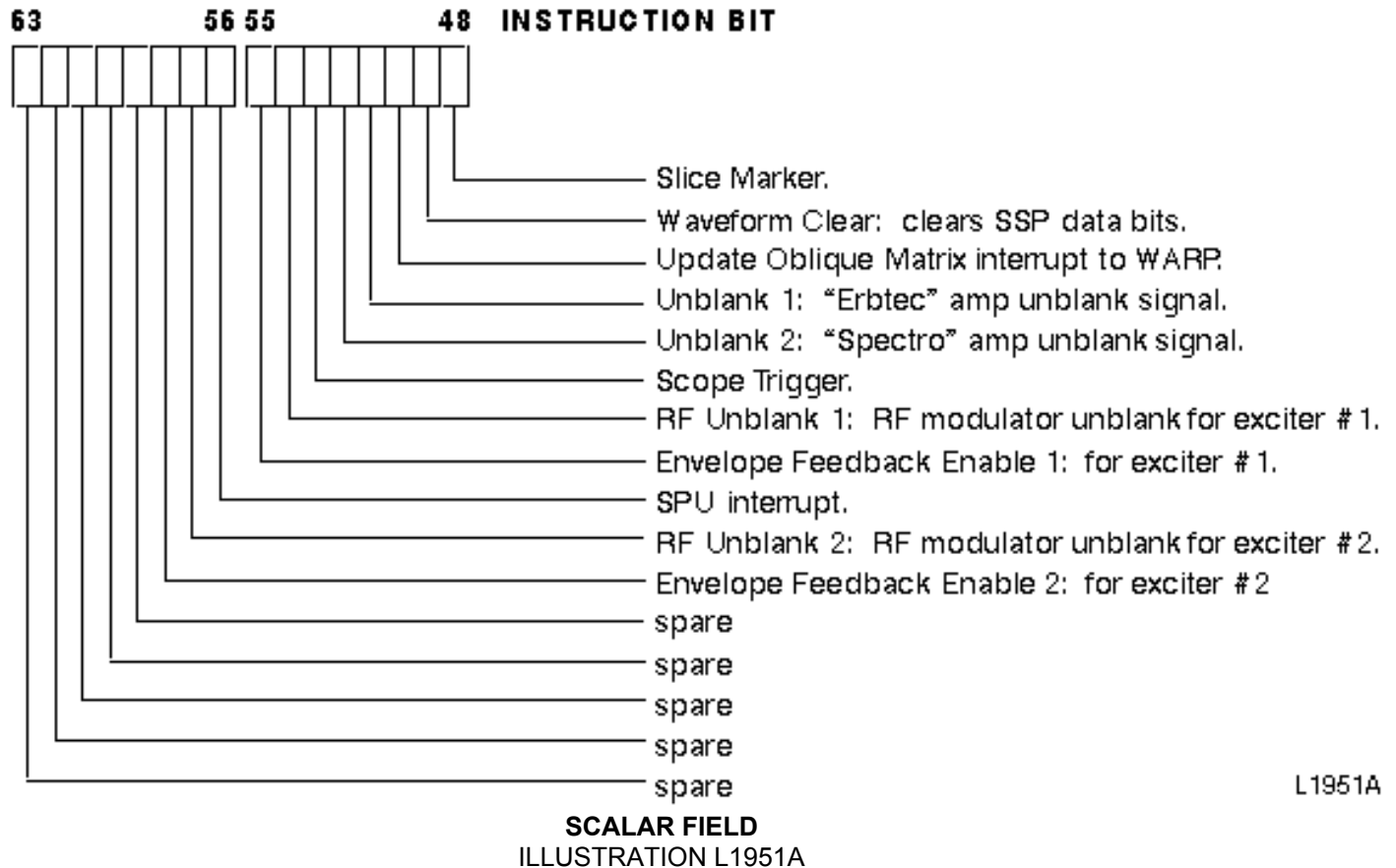
bit 31: The cine bit (SSP only) is also latched externally for the 68020 to read.

6- THE SCALAR

The scalar in the sequencer instruction is a 16-bit, two's complement number that is multiplied with the 15-bit, two's complement waveform data. The same scalar value is multiplied with each of the waveform points played out within a single instruction.

The scalar is present only for the gradient and RF sequencers. The SSP generator sends its waveform data directly to the backplane, and has no need to scale those data.

The scalar field for the SSP generator contains unique, real time, control bits. See Illustration L1951A.



7- WAVEFORM MEMORY

The waveform memory is 256K by 16 bits. All of the sequencers share this memory. The format of waveform memory varies, depending on the sequencer that is using it. It is accessed as words only by either the 68020 or the sequencers. It is mapped at address 00C00000 to 00C7FFFE for the 68020 and address 00000 for the sequencers.

The three gradient and four RF sequence generators have the same bit definition. It is as follows:

bit 0: End of waveform (EOW) bit

bits 1–15: Two's complement waveform value (15 bits)

The SSP generator bit definition is:

bit 0: SSP Data 0	bit 10: unused
bit 1: SSP Data 1	bit 11: unused
bit 2: SSP Data 2	bit 12: unused
bit 3: SSP Data 3	bit 13: unused
bit 4: SSP Data 4	bit 14: unused
bit 5: SSP Data 5	bit 15: End of
bit 6: SSP Data 6	waveform (EOW)
bit 7: SSP Data 7	
bit 8: SSP Tag 0	
bit 9: SSP Tag 1	

8- X, Y, AND Z ADDRESS LATCHES

The x, y, and z gradient sequence generators share one set of address latches. They can do this since none of their instruction or waveform memory access cycles overlap. The sequence generator LCAs individually multiplex their output between instruction memory and waveform memory addresses. The LCAs drive the bus only during their defined time slots, and go tri-state for the remainder of the time.

9- THE X, Y, AND Z WAVEFORM AND SCALAR LATCHES

The scalar latch is loaded at the end of the second instruction memory cycle for a given sequence generator and the waveform latch is loaded at the end of the waveform memory cycle. Each latch is read by the WARP processor which provides the necessary READ signals to the _RD pins. When active, the _RD pin causes the Q outputs to turn on.

10- THE SSP SEQUENCE GENERATOR

Since the SSP generator is not mutually exclusive in time with any of the other sequencers, it must have unique waveform and instruction memory address latches. These latches operate in the same manner as the x, y, and z gradient latches described above with the exception that the SSP_IOUT and SSP_WOUT signals never go tri-state.

The SSP generator latches its instruction control bits (instruction control field and scalar field) externally. Some can be read by the 68020, and others are driven out over the backplane. The waveform data are latched in the SSP Bus Output Registers, and are strobed over the backplane with the 400-nsec, active low SSPSTB. The four UNBLANK signals are buffered and accessible on SMB connectors on the foreplane.

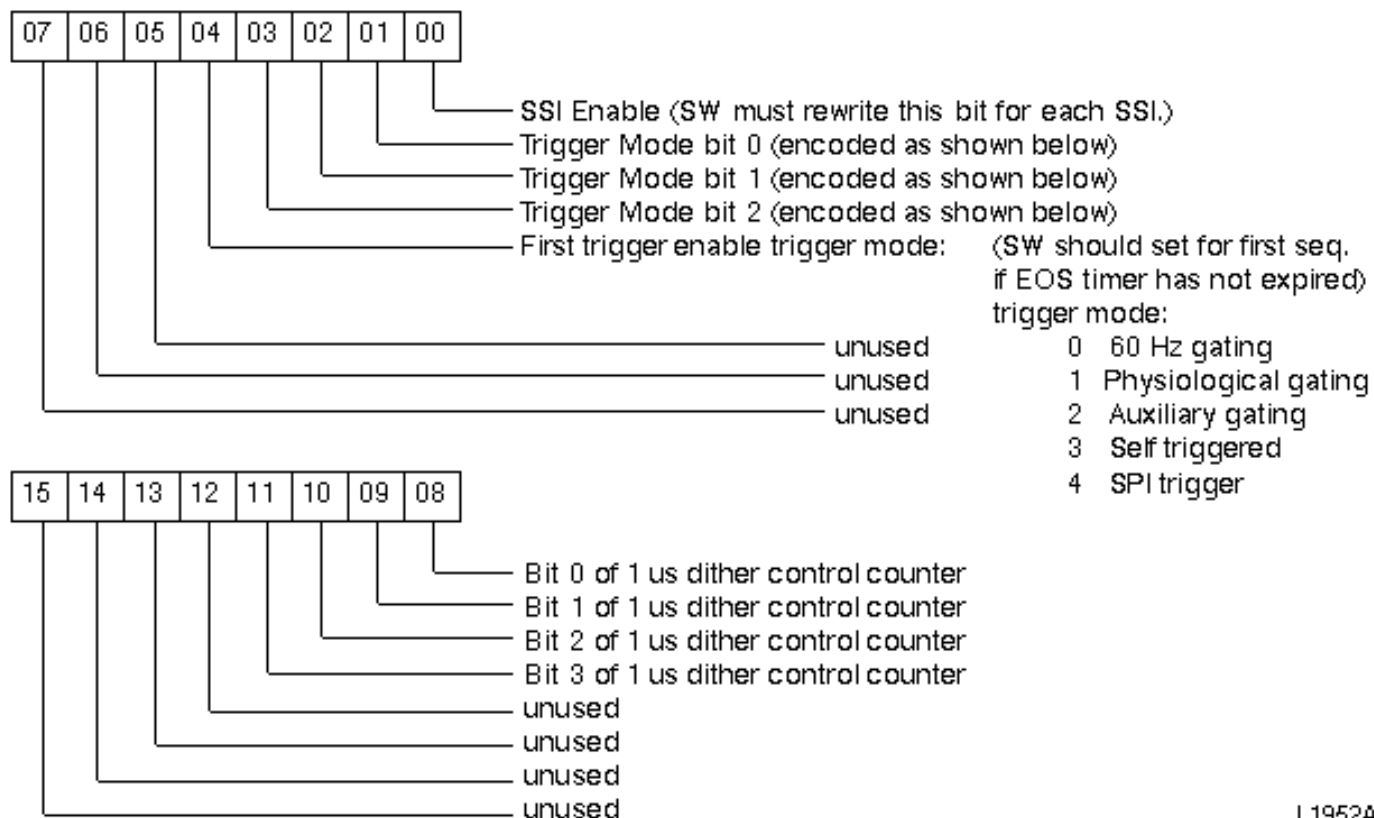
11- THE MASTER SEQUENCER

The master sequencer is an LCA that controls the rest of the sequence generators and all of the bus arbitration needed by them for waveform and instruction memory access. The master sequencer also contains a control register that defines the triggering mode and enables a sequence to start. The sequence of events for sequencing is as follows:

1. The IPG 68020 sets the SSI enable bit within the master LCA. A sequence begins and a start of sequence interrupt (SSI) is generated by the master sequencer. Then all of the slave sequencers are started up in a staggered manner by the master sequencer.
2. Instructions are played out by each sequencer until each receives its own end of sequence (EOS) bit. The SSP sequence generator must be the last sequencer to see an EOS bit and must play at least four microseconds longer than any of the other sequencers. This allows the pipeline associated with the WARP and the RF multiplier control to be emptied prior to the complete shutdown of the sequencers.
3. Once the SSP generator executes an instruction with an EOS bit set, the master sequencer immediately disables all of the slave sequence generators by pulling all of their enable lines low. The SSPEOS bit remains set until the next instruction is played.
4. An EOS timer within the master sequencer LCA is started when an SSP EOS is detected. This timer is a 16-bit timer with a 4- μ sec clock driving it, yielding a maximum EOS time of 256 milliseconds.

12- THE MASTER SEQUENCER CONTROL REGISTER

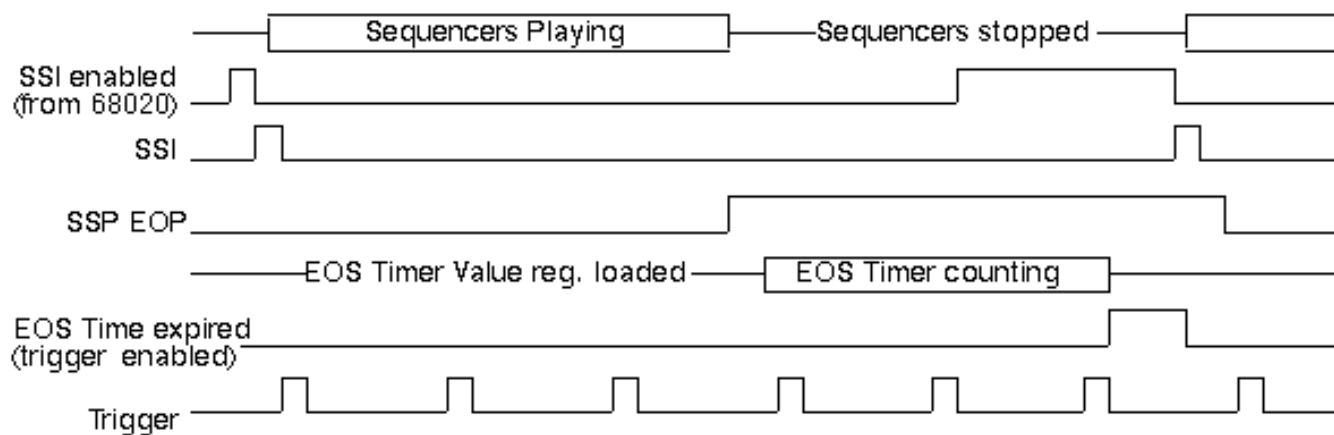
The master sequencer control register, which is written by the IPG 68020, is defined by Illustration L1952A and Illustration L1953A.



MASTER SEQUENCER CONTROL REGISTER
ILLUSTRATION L1952A

L1952A

"Triggering" Timing diagram



L1953A

"TRIGGERING" TIMING DIAGRAM
ILLUSTRATION L1953A

13- SSI ENABLE

The SSI enable circuit is integrally related to the triggering operation described in the above timing diagram and control register bit definition. The first SSI enable for a given scan must have bit 4 of the control register set, as well as the normal SSI enable bit. This requirement results from the implementation of the control logic associated with the trigger. The bit 4 enable essentially bypasses the triggering mechanism.

The SSI logic requires that the SSI enable bit be set (bit 0), and that the EOS timer has expired before a trigger is recognized. SSI occurs first as shown in the above timing diagram. The EOS timer would not have even started for the very first sequence of a scan; therefore, the first trigger enable (bit 4) is provided to kickoff the first sequence. From then on, the SSI logic looks for a normal SSI enable (bit 0), an expired EOS timer, and a trigger.

It is important to note that even after the 68020 sets bit 0, the trigger signal is not latched up until after the EOS timer expires. The above timing illustrates this point by showing a trigger after SSI enable yet before the EOS timer has expired. SSI occurs only after both events have occurred. It is also important to note that the 68020 should always set the SSI enable bit before the EOS timer expires.

14- THE EOS TIMER

The EOS timer value register (Address 00CA0002) is 16 bits. It contains the two's complement number representing the number of 4- μ sec periods that are allocated for the EOS dead time, i.e., the time between the EOS of one sequence and the SSI of the next sequence.

The actual EOS time is extended by 8 μ sec from the number written into the EOS timer value register as a result of the internal operation of moving the value from the register to the timer. It would also be delayed by the trigger itself which could vary depending upon what type of trigger it is.

The EOS counter bits themselves are at the 68020 address 00CA0004, with the exception of bit 0 which is the EOS timer expired bit. It is to be read by the 68020 and verified not to be set prior to rewriting the SSI enable bit. When both the SSI enable bit and the terminal count of the EOS timer occur, the trigger circuitry is enabled allowing for a new SSI to be generated.

15- TRIGGERING MODES

Five triggering (gating) modes are defined above in the master sequencer control register section. The 60-Hz, Physiological, Auxiliary, and SPI triggers all operate as described in the above SSI enable and EOS timer sections. The self-triggering mode acts slightly different. The self trigger is actually the EOS timer Expired bit. In that case, SSI occurs immediately after the EOS timer has expired provided the SSI enable bit has been set.

16- TR DITHER CONTROL

Four bits are defined in the master sequencer control register that control the TR dither. These four bits give 0 to 15 μ sec, in 1- μ sec steps, of additional time from one SSI to the next SSI. The time is added at the beginning of EOS time (just after the SSP generator has executed an EOS).

17- WARP SYNCHRONIZATION AND PIPELINE DELAY

The WARP (TMS320C30) is kept synchronous with the sequence generators by the master sequencer through a WARP interlocked load instruction. The WARP attempts to read the x gradient scalar and waveform registers with an interlocked instruction. The processor generates an XF0* in doing so. For the cycle to complete, the WARP must see a low on its XF1 pin. When the master sequencer sees the XF0, it provides an XF1_INTLK* signal beginning at the second instruction cycle of the y gradient sequencer. The XF1_INTLK* signal remains asserted until the WARP removes XF0.

The WARP reads the y and z gradient data without the interlocked instruction. It executes a fixed number of instructions between the reading of the x, y, and z gradient sequencers scalar and waveform registers to keep it in sync. The time between the three, double reads is precisely 1 μ sec. The XF0 / XF1 handshake occurs every 4 μ sec. The WARP reads the data, performs a rotation algorithm, and writes it out to some shift registers at that 4- μ sec rate. This causes a pipeline delay with respect to the SSP sequence generator that has its data played out directly to the backplane. An additional delay, defined in the WARP section of this document, is present because the gradient data is shifted out serially to the gradient amplifier.

18- THE UPDATE MATRIX INTERRUPTS

The master sequencer queues up and synchronizes the two interrupts that tell the WARP processor to read new matrix coefficients for its rotation algorithm. The interrupts come from either the LCA control register which is written by the 68020 or from an SSP sequencer control bit.

The master sequencer prevents XF1_INTLK*s from being generated when it receives either of the update matrix interrupts. It sends the queued interrupt to the WARP at the same time that it would have sent an XF1_INTLK*. After the WARP has processed the interrupt, it clears the XF1_INTLK* hold off circuitry within the master LCA. Normal XF0 / XF1 handshakes then resume.

19- ISA WRITE CONTROL PALS

The instruction start address (ISA) write control PALS create the load ISA register (LIR) strobes for each of the slave sequencers. The write strobes are synchronized with the time slot associated with the 68020 access to the instruction memory banks. That way, the tri-state transceivers normally used by the 68020 for instruction memory access can also be used to get into the LCA of the slave sequencers.

20- LCA CONTROL AND STATUS

The LCAs are programmed in a slave serial mode. The LCA control PAL generates the configuration clock required to load the bit streams into the LCAs. Ten unique serial bit streams are mapped to ten unique bits of the 68020 at address 00CC0000 as follows:

Bit: LCA function:

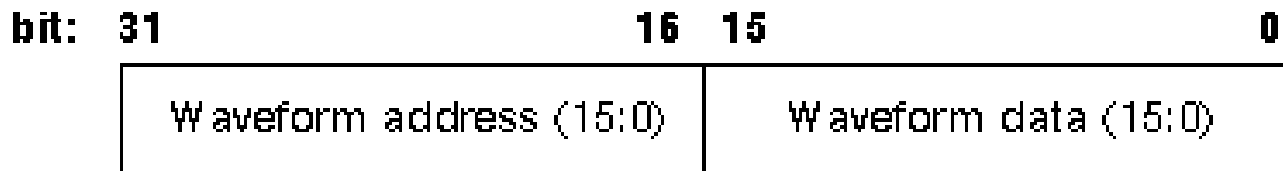
- 0 SSP sequence generator
- 1 X gradient sequence generator
- 2 Y gradient sequence generator
- 3 Z gradient sequence generator
- 4 WARP gradient waveform output
- 5 Rho 1 (RF) sequence generator
- 6 Rho 2 (RF) sequence generator
- 7 Theta 1 (RF) sequence generator
- 8 Theta 2 (RF) sequence generator
- 9 Master sequencer

The bit streams can be read back at address 00CC0002 with the same bit assignment for verification. A DONE-PROG* signal is provided in the LCA status register at 68020 address 00CC0006 to indicate that all of the LCAs have been downloaded. Also provided in the status register is a board ID that can identify sixteen different versions of artwork. The LCA Control Register (address 00CC0004) contains the control bits necessary for programming the LCAs. It also contains the update matrix interrupt bit for the WARP processor and the two most significant waveform memory address bits needed during the fast update described next.

21- WAVEFORM MEMORY FAST UPDATE

A special technique for addressing the memory is provided. It's called *fast update*, and it takes a 32-bit data write from the 68020. It uses the upper 16 bits as address, and the lower 16 bits as data. The 32-bit value is always written to address 00CB0000 to write to waveform memory. This allows the 68020 to prepare a file in non-critical times, and then just dump the file to address 00CB0000. The actual memory locations being modified can then be anywhere in waveform memory. This fast update feature is only for writing to memory.

The waveform memory requires eighteen address bits. Two page bits in the LCA control register are used in conjunction with the upper 16 bits to form the 18-bit address. The 18-bit address is a "word" address, i.e., it addresses 256K, 16-bit locations. See Illustration L1954A.



WAVE_PAGE_0 = waveform address 16

WAVE_PAGE_1 = waveform address 17

L1954A

LCA CONTROL REGISTER
ILLUSTRATION L1954A

The waveform address as defined above is the "absolute" address of 16-bit locations in waveform memory with no 68020 address offset.

The control logic causes the 68020 to wait for its assigned time slot. This causes a maximum update rate of 1 mHz, or once per microsecond. Since the control of the time slots is done by the master sequencer, the LCAs must be downloaded prior to writing to the waveform memory.

22- NORMAL 68020 R/W PATH TO WAVEFORM MEMORY

The 68020 is granted access to the waveform memory only once per microsecond. The LCAs must be downloaded prior to any attempted waveform memory access since the time slots are controlled by the master sequencer. The control logic queues up the 68020 access, allowing it to complete when the WAVE_020_EN* signal goes low, at which time the address and data busses are connected.

23- INSTRUCTION MEMORY (BANK #1) FAST UPDATE

The fast update to the instruction memory works nearly identically to that of waveform memory. The only difference is the absence of the page bits and the most significant bit in the instruction address field is a "don't care". That is because one instruction memory bank is only 64 K, 16-bit locations (32 K, 32 bit locations) requiring only 15 bits of address. The 68020 writes to address 00CB0008 to perform a fast update to instruction memory bank #1.

24- NORMAL 68020 R/W PATH TO INSTRUCTION MEMORY BANK #1

The 68020 can access the instruction memory as words or long words (16 bits or 32 bits, respectively). It is mapped at address 00D60000 through 00D7FFFE. The control logic in the Instruction Control PAL causes the 68020 to wait until its appropriate time slot as defined in the sequence generator memory access map.

25- RF BUS CONTROL AND SEQUENCE GENERATORS

The RF section contains three lower blocks: sequencers, multiplier, and memory. The RHO and THETA address latches are similar to the x, y, z address latches. The following sections describe the multiplier and memory blocks.

26- RF MULTIPLIER AND DATA REGISTERS

The RF multiplier subsection controls the multiplier and its registers, as well as the RHO bus and the THETA bus.

As the RF sequence generators supply data to the CY7C516 multiplier, the RF control latches the product into one of four multiplier registers. The four multiplier registers correspond directly to the four sequence generators, RHO1, RHO2, THETA1 and THETA2.

As stated previously, each RF generator can provide new values every two microseconds. Any latched RHO or THETA data is played out on the corresponding bus every microsecond. So when all four sequencers are running, the data from RHO1 and THETA1 registers are sent out simultaneously; and the data from the RHO2 and THETA2 registers are sent out one microsecond later.

In addition to the RHO bus and THETA bus, the RF section drives four strobe signals and a frequency select line on the backplane. When the RHO data bus is driven, RHOSTB1* or RHOSTB2* indicates which exciter board is to receive the data. The two RHO strobe signals directly correspond to the RHO1 and RHO2 multiplier registers and sequence generators.

Similarly, when the THETA bus is driven, THTSTB1* or THTSTB2* indicates which exciter board is to receive the data. However, the THETA strobes do not necessarily correspond directly to the THETA1 and THETA2 multiplier registers. The THETA1 sequence generator produces either phase for exciter 1 (THETA1) or frequency for exciter 2 (OMEGA2). The THETA2 sequence generator produces either phase for exciter 2 (THETA2) or frequency for exciter 1 (OMEGA1). When frequency data are generated and sent on the THETA bus, the frequency select line, FREQSEL, is driven high. Therefore, when frequency data for exciter 1 are sent, THTSTB1* and FREQSEL are active, and the THETA bus is driven with data from the THETA2 multiplier register.

27- INSTRUCTION MEMORY BANK #2 AND SCALAR LATCHES

Instruction memory bank #2 can be accessed via a normal 68020 R/W mode, or a fast update mode similar to accesses of instruction memory bank #1. The bank #2 normal R/W accesses are mapped from address 00D80000 through 00D9FFFE. The 68020 writes to address 00CB00C to perform a fast update to instruction memory bank #2.

The MAX PAL internally contains four 16-bit latches for RHO1, RHO2, THETA1, and THETA2 (or OMEGA) scalar data. Each RF sequence generator sends a scalar enable signal (ENR1*, ENR2*, ENTH*, ENOM*) which enables the corresponding MAX PAL register onto the SCALE bus. The MAX PAL also generates the signals for latching data into and out of the multiplier (MINCLK and MOUTCLK). The other outputs of the MAX PAL (RHO1STB, RHO2STB, THT1STB, and THT2STB) are used for latching the multiplier output into the multiplier registers.

28- WAVEFORM AND ROTATION PROCESSOR (WARP)

The Waveform and Rotation Processor (WARP) section of the IPG board uses a TMS320C30 microprocessor to perform gradient scaler and waveform multiplication, and to perform matrix multiplication of the gradient values during oblique imaging. The resultant gradient data are serialized, and are sent over the backplane to fiber optic transmitters on the TYME board, and out to the gradient amplifiers.

29- DUAL-PORT RAM

The WARP dual-port RAM consists of 2K x 32-bit memory. The WARP CPU (TMS320C30) always accesses this memory as 32 bit, beginning at address 0H. The MC68020 can access this memory as word (16 bit) or long word (32 bit), beginning at address CD0000H.

30- BOOTING

The WARP CPU is held reset by the MC68020 while boot code is downloaded to the WARP dual port RAM by the MC68020. Once the WARP CPU reset signal is released, the TMS320C30 executes code from the dual port RAM.

31- SEQUENCER INTERFACE

Waveform and scaler data for each of the three gradient axis are latched into registers by the gradient sequencers. The WARP CPU reads (zero wait states) the sign extended data from these registers through its expansion bus. The WARP CPU is pseudo-synchronized with the sequencers via the XF1 and XF0 pins, and an interlocked load instruction. The use of the interlock pins in this application follows CPUs:

1. The WARP CPU, after processing data for the previous set of gradient points, simultaneously resets XF0 to 0 and starts a read cycle of the X scaler register. This read cycle is held off until the TMS320C30 sees XF1 go low.
2. When the master sequencer is ready to have the WARP CPU read the waveform and scaler registers, and process the data, it resets XF1 low, allowing the WARP CPU to complete the interlocked read cycle, perform noninterlocked reads of the other waveform and scaler registers, and process the data.

32- OBLIQUE PLANE IMAGING

The WARP CPU performs matrix multiplication of the gradient data with matrix coefficients previously stored into the WARP dual-port RAM by the MC68020. An example of events that may be used by the WARP processor follows:

1. The gradient waveform and scaler data are read from the gradient sequencer registers as indicated in section 31-Sequencer Interface above.
2. In addition to the scaler/waveform multiplication, matrix multiplication is performed on the resultant gradient data and matrix coefficients previously read by the WARP CPU from the dual WARP dual-port RAM.
3. New matrix coefficient values are written to the WARP dual-port RAM by the MC68020. At the correct time, the master sequencer asserts the WARP CPUs update done interrupt (INT0), telling it to update the matrix coefficient values it has in internal RAM. After reading the new values from dual-port RAM, the WARP CPU asserts the UPDATE_DONE signal, notifying the master sequencer that the coefficients have been updated.

33- PARALLEL-TO-SERIAL CONVERTER

The least significant 20 bits of the post-processed x, y, and z gradient data (in this order) residing in the WARP CPU are written to 20-bit storage registers in a Xilinx XC3042 LCA. After the z gradient data are written to the LCA register, followed with XF1 going low, the data from all three LCA gradient storage registers are latched into 20-bit shift registers, and shifted out at a 10-mHz rate. In addition to the three gradient data signals leaving the LCA, there is a clock signal that is sent to the gradients to synchronize the data transfer. All four signals are sent through J2 to the TYME board where the signals are inverted and sent over fiber optics to the gradient amplifiers. The three LCA gradient storage registers can be read back by the WARP CPU for integrity checks.

34- SPU REGISTER

There are three 10-bit registers in the Xilinx LCA that contain the 10 most significant bits of the same gradient data that are latched into the LCA gradient registers. The data from these registers can be read by the SPU CPU (TMS320C30) to get the gradient data it requires to filter out gradient noise from patient monitoring data. Because the same internal LCA bus is used for both the WARP and SPU registers, the SPU cannot read its registers while the WARP has access to the LCA. The WARP CPU is given access to the LCA registers whenever it needs them by arbitration logic internal to the LCA. The WARPs L_S_LOCK* signal was implemented as an indicator to the SPU as to when it can access the LCA registers.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.